

BOARD OF INTERMEDIATE EDUCATION KARACHI

HSC PART II · CLASS XII

Computer Science Lecture Notes

Worked examples, board-style practice, and practicals

Session 2026–27

Sindh Textbook Board chapter scheme

New Sindh Curriculum of Computer Science 2024

Aligned with the National Curriculum of Pakistan 2022–23

Theory 75 marks · Practical 25 marks

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How to use these lectures

These notes are for classroom teaching of Computer Science at the Board of Intermediate Education Karachi, Paper II, Class XII. They follow the six units of the Sindh Textbook Board Class XII book under the New Sindh Curriculum of Computer Science 2024, aligned with the National Curriculum of Pakistan 2022–23. They assume the Class XI lectures: gates, searches, Python functions, a first database, and a prototype. They are original teaching notes, not an official board publication and not a substitute for the prescribed textbook or the year’s model paper.

How the book is arranged

Lecture	Textbook unit	You will be asked to
1	Computer Systems	Judge usability, accessibility, and security together
2	Computational Thinking and Algorithms	Use stacks, queues, and trees, and evaluate a solution
3	Programming Fundamentals	Write a class, use files and dictionaries, and normalise to 3NF
4	Data and Analysis	Measure a model and write a hypothesis test without over-claiming
5	Applications and Impacts	Design a Pakistani application and a privacy compromise
6	Entrepreneurship	Test a minimum viable product against a failure rule

The Class XII book.

Marks

Paper II is 75 marks of theory and 25 marks of practical, as in the BIEK scheme of studies. Practise multiple choice, short answers, and long answers, and confirm the current model paper’s exact split. Several Class XII questions are two-part: a calculation or a traversal, then a sentence that limits the claim. Leave time for the sentence. It is often the difference between a mechanical answer and full marks.

How to write an answer

- For a tree, write the rule (inorder, preorder, postorder, BFS, or DFS) before the list of nodes. Use the diagram printed in the question.
- For a metric, write the formula, then the substitution, then the percent. Say what the percent counts.
- For a hypothesis, write H_0 and H_1 in the words of the study. Do not invent a p-value the question did not give.
- For a design, name the user, the data, the technology you refused, and one risk.
- For an MVP, write the failure rule. An answer with no way to fail is a poster, not a test.

CLASS XII · UNIT 1

Lecture 1

Usable, Accessible, and Secure Systems

Textbook unit. Unit 1 — Computer Systems

Curriculum domain. A. Computer systems

Class time. about 30 periods

Class XI asked how a system is built and how bits move. Class XII asks whether a person can actually use what was built, whether someone who sees, hears, or moves differently can use it, and whether the security added along the way makes the system so awkward that people route around it. Human-computer interaction is the name of that design work.

Learning outcomes

- Explain usability, accessibility, and security, and give an example of each failing on a familiar system.
- Describe human-computer interaction in terms of usability, common problems, improvements, and ethical, social, economic, and environmental effects.
- Explain a usability test in enough detail to carry one out on a paper screen.
- Recommend a security measure that respects efficiency, cost, privacy, and ethics, including the idea of a zero-trust design.

Three words that are not the same

Usability is how effectively, efficiently, and satisfactorily a stated user can complete a stated task. Effectively means they can finish it. Efficiently means the effort is reasonable. Satisfactorily means the experience is not a punishment. **Accessibility** means people with disabilities, and people in hard conditions, can complete the same task. **Security** means the system resists harm to confidentiality, integrity, and availability. A door that nobody can open is secure and useless. A door with no lock is usable and unsafe. The design work is the space between those jokes.

HUMAN-COMPUTER INTERACTION

Human-computer interaction (HCI) studies how people use computing systems and how to design those systems so the use is effective, efficient, and satisfying. It includes the layout of a screen, the words on a button, the time a task takes, the errors people make, and the consequences when the design ignores a group of users.

Usability: problems and improvements

Common problems are dull and expensive. The label does not match the user's word ("submit" when the user thinks "save"). There is no confirmation, so people press twice. The error message says "invalid" and not what to fix. Text is grey on a pale background. The only way forward is a tiny link. A long form asks for a father's name when the task is to see a date sheet. Each of these is a usability fault even if the database behind it is correct.

Improvements are equally concrete. Use the user's words. Put the primary action where the eye arrives, and make it look like a button. Keep a consistent place for navigation. Show the password rules before the user fails them. Let the user undo. Test with five people who are not the programmer and watch where they stop. A **usability test** gives a person a goal, stays quiet, and records the stops. It is not a demo in which you drive the mouse while they watch.

WORKED EXAMPLE — A COLLEGE PORTAL TASK

Task: a student must download this week's CS practical sheet.

On the current portal the sheet is under “Academics Resources Misc files (3)”. Three test users look under “Computer Science” and give up. That is a usability finding, not a user failure.

Improvement: a heading “This week” on the CS page, the practical's name in the link, and the upload date. Retest with two new students. If they reach the file without help, the change did the job. If the file is a 40 MB scan of a photocopy, you have found a second problem.

Accessibility

An inaccessible system excludes people the institution has already accepted. A screen-reader user needs real text, not a photograph of text, and needs every control to be reachable from the keyboard. A user with low vision needs contrast and the right to enlarge text without the layout collapsing. A deaf user needs captions on a teaching video. A user with a tremor needs a target that is not a few pixels wide. Colour must not be the only signal that a mark is failing; a word or an icon has to carry it too, because some users do not distinguish red and green.

The effects of ignoring this are not limited to one annoyed person. A student misses the date sheet and the admission. A public service that only works in one language, on one expensive phone, teaches the public that the service is not for them. Accessibility overlaps the digital divide from Class XI. In Class XII you are expected to talk about the design, not only about the gap.

Effects of a bad design

Kind	A bad result	A design response
Ethical	People are tricked by a button that does more than it says	The label matches the action; consent is specific
Social	A group cannot use the service and drops out of the activity	Language, captions, and a path that works on a small screen
Economic	Staff spend the day resetting passwords and retyping forms	Clear errors and a cheaper support load
Environmental	Users print every page because the screen version is unreadable	A readable page and a single clean print

Table 11. Use this shape when a question lists ethical, social, economic, and environmental implications.

Security that people will actually follow

There is a standing **tradeoff** between security and usability. A password of 30 random characters is harder to guess and harder to remember, so people write it on a sticky note, and the sticky note may be the weakest point. A session that expires every two minutes protects a shared lab PC and infuriates a teacher entering marks. The designer's job is to say which risk is real in this setting and to pick a control that users will not bypass.

The factors to name are **efficiency** (does the task still finish in a reasonable time?), **cost** (can the college pay for the control, and what does a breach cost?), **privacy** (are you collecting more data than the control needs?), and **ethics** (are you watching people who were not told, or locking out people who cannot use the chosen method?). Biometrics, for example, can be fast and can also be a problem for a student whose fingerprint is worn from work, and a fingerprint cannot be changed if the database leaks. A password can be changed. Say that limit if you recommend a fingerprint.

A **zero-trust** approach means the network does not treat a device as safe merely because it is inside the building. Every request is authenticated and given only the access that request needs. In a college, that sounds like: the lab PC is not automatically allowed to open the result database; the

clerk's account can enter marks for this class and not download the whole board's rolls; a forgotten laptop does not carry a permanent login. Zero trust is a design habit, not a product you buy and forget.

WORKED EXAMPLE — A USABLE CONTROL

Risk: students reuse one weak password on the portal.

A poster that says “use a strong password” does almost nothing. A usable design lets the student paste a password from a password manager, shows the rules before submission, and turns on a second factor that the student already has, such as a one-time code, with a recovery method that does not depend on a single lost phone. Forcing a change every week looks strict and trains people to use Password2, Password3, and a notebook. That fails both usability and security.

Cost: the code-by-SMS path costs money and fails when the network fails, so a code from an authenticator app, plus a printed recovery code kept at home, may fit better. Privacy: the portal does not need the student's contacts in order to protect the login.

COMMON MISTAKE

“Add more passwords” is not an HCI answer. Name the user, the task, the risk, the control, and the way the control could itself be bypassed.

Key terms

Term	Meaning in this lecture
Usability	Effectiveness, efficiency, and satisfaction for a stated user and task.
Accessibility	The same task can be completed by people with disabilities and in hard conditions.
HCI	The study and practice of designing that use.
Usability test	A user attempts a goal while the designer watches and does not help.
Tradeoff	A gain in security that costs usability, money, privacy, or fairness.
Zero trust	Every request is checked; being on the local network is not enough.
Least privilege	An account can do only what its job needs.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Why is a usability test not a demonstration?

MODEL ANSWER

In a demonstration the designer operates the system and the story succeeds. In a test the user operates it, the designer stays quiet, and the stops are the result.

Give one accessibility fault that is not about blindness.

MODEL ANSWER

A video with no captions excludes a deaf student and also fails a student in a noisy room. A button that is too small excludes someone with a tremor.

Why can a very strict password rule reduce security?

MODEL ANSWER

People who cannot remember it write it down or simplify it in a predictable series. The rule is bypassed, and the bypass may be weaker than a memorable long passphrase would have been.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Which situation is a usability-security tradeoff?

- A. A short password that is easy and easy to guess
- B. A faster processor
- C. A larger monitor that shows the same form
- D. A primary key

2. A usability test's main evidence is

- A. the designer's confidence
- B. where a real user stops or succeeds
- C. the number of colours
- D. the price of the server

3. Zero trust means

- A. nobody is allowed to log in
- B. a device on the office network is still checked on each request
- C. passwords are abolished
- D. users are not told the rules

4. Using red alone to mark a failing grade fails

- A. only the database
- B. accessibility, because colour must not be the only signal
- C. binary search
- D. the mean

Answer key with reasons

Q	Answer	Why
1	A	Ease and resistance to guessing pull in opposite directions. The other options are not that conflict.
2	B	Watching the user's attempt is the test. Confidence and decoration are not evidence.
3	B	Location inside the building is not treated as proof. Access is checked and limited.
4	B	Some users cannot distinguish the colour, so the word or an icon must carry the meaning too.

Short questions

S1. Define usability using the three standard parts, and illustrate one part with a portal task.

MODEL ANSWER

Usability is effectiveness, efficiency, and satisfaction for a particular user and task. Effectiveness: the student can download the practical. Efficiency: they do it in a few steps, not a tour of the site. Satisfaction: the labels match what they expect, so the task is not frustrating. A test watches them try.

S2. Give one economic and one environmental effect of an unreadable on-screen notice.

MODEL ANSWER

Staff reprint and re-explain the notice, which costs time. Users print it themselves because they cannot read it on the phone, which wastes paper and toner. A readable page with a single official print reduces both.

S3. Recommend one security measure for a shared lab PC and say how you keep it usable.

MODEL ANSWER

End the session when the browser closes or after a short idle time, so the next student does not inherit the account. Keep the idle time long enough to read a question, and put a visible countdown, so people are not thrown out mid-sentence. Personal passwords on a shared desktop profile are the wrong control.

Long questions

L1. (5 marks) Explain human-computer interaction for a result portal under the headings usability, common problems, methods of improvement, ethical and social impact, and economic impact.

MODEL ANSWER

Usability: a student should open their own result in a few steps and understand every label.

Common problems: the roll number is asked for in a format the admit card does not use; the error says only “invalid”; the result is a scanned image that a screen reader cannot read.

Improvement: show an example of the roll-number format, name the error, publish HTML or text as well as a print file, and run a usability test with students who did not build the site.

Ethical and social: publishing a full list in roll-number order on an open wall may shame students and exposes data the task does not require. Showing a student their own result after login is the tighter design. An inaccessible scan excludes a blind student from the same news everyone else has.

Economic: a confusing portal fills the office with visitors who only needed a file, so the college pays in staff time. Clear self-service is the cheaper system after the first careful design.

L2. (5 marks) A bank app wants both a fast login and protection against a stolen phone. Describe the tradeoff and a recommendation that mentions cost, privacy, and a user who might be locked out.

MODEL ANSWER

A login that is only a swipe is fast and weak if the phone is unlocked. A login that demands a long password plus two extra codes is stronger and will be bypassed or abandoned. The tradeoff is real: every extra step stops some thieves and also stops some customers.

A fitting recommendation is a device unlock the user already has, then a second factor for payments only, not for looking at the balance. Cost stays down if the second factor is an authenticator or a prompt rather than an SMS on every glance. Privacy: the app does not need the contact list to do this. Ethics and lockout: a customer whose fingerprint fails, or who has lost the phone, needs a recovery path that a branch or a pre-printed code can complete, or the security measure becomes a refusal of service. Zero trust shows up as the payment being checked again even though the balance screen was already open.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 1 — Watch a user, then recommend a control

- 1 Sketch three paper screens of a task you know (finding a practical, or logging into a lab account).
- 2 Give the paper to a classmate. Set the goal. Do not help. Note every hesitation.
- 3 Change one screen. Retest with a different classmate.
- 4 Add one security control to the same task. Write who might be locked out by it and how they recover.

WHAT SUCCESS LOOKS LIKE

The journal has the two test notes, the change, and a security recommendation that names a recovery path.

Copy this summary

- Usability is effectiveness, efficiency, and satisfaction. Accessibility widens who can succeed. Security limits harm.
- HCI studies that use and designs for it.
- A usability test watches a user attempt a goal. It is not a demo.
- Bad design has ethical, social, economic, and environmental effects. Give one concrete effect, not four adjectives.
- Security and usability trade off. Name efficiency, cost, privacy, and ethics.
- Zero trust checks each request. Least privilege limits what the successful login may do.
- A control without a recovery path locks people out and will be bypassed.

CLASS XII · UNIT 2

Lecture 2

Data Structures, Trees, and Evaluation

Textbook unit. Unit 2 — Computational Thinking and Algorithms

Curriculum domain. B. Computational thinking and algorithms

Class time. about 20 periods

Class XI traced searches and sorts on a list. Class XII asks you to choose a structure, walk a tree in a stated order, and judge a solution by efficiency, clarity, and correctness. The structures are the list, the array, the stack, the queue, and the tree, with binary search on a sorted collection.

Learning outcomes

- Evaluate an algorithm for efficiency, clarity, and correctness using a specific example.
- Describe lists, arrays, stacks, and queues and name a fitting use for each.
- Draw a binary search tree and produce inorder, preorder, and postorder traversals.
- Explain why binary search needs sorted data and how a tree supports that search.
- Show the visit order of a breadth-first and a depth-first walk on a small tree.

Judging a solution

Three questions are enough. **Correctness:** does it meet the specification on the awkward cases, not only on the example in the notes? **Clarity:** can another student trace it, with names and structure that match the problem? **Efficiency:** how does the work grow as the data grows? At this level, compare “one pass through n items” with “about one comparison for each time you can halve n ” or with “every item compared with every other item”. You may write $O(n)$ and $O(\log n)$ if you have been taught the notation. A sentence in ordinary English scores the mark if it is accurate.

WORKED EXAMPLE — JUDGING A SEARCH

Specification: find a roll number in a list of 5,000 sorted roll numbers.

Linear search is correct and clear, and it may look at all 5,000. Binary search is correct only because the list is sorted, it is clear if low and high are updated in the right direction, and it finishes in a handful of comparisons because each step discards half of what remains. Binary search on an unsorted list is not “less efficient”. It is incorrect. A clever one-line version with no names is less clear and can lose marks even when it is right.

Lists, arrays, stacks, and queues

A **list** is an ordered collection that can grow and shrink. You have used Python lists. An **array** is a fixed block of same-typed cells addressed by index. In exams, “array” often means the idea of indexed storage, while “list” means a sequence you can append to. The practical difference you must be able to say: finding an item by index is direct; finding it by value may require a search; inserting in the middle of an array shifts the later items.

A **stack** is last-in, first-out. The operations are push and pop. The call stack of a program is the serious example: the most recently called function returns first. Undo in an editor is a stack of actions. You do not pull an item from the middle.

A **queue** is first-in, first-out. Enqueue at the back, dequeue at the front. A printer queue, a lab signup, and the waiting list for breadth-first search are queues. A stack would print the most recent file first, which is the wrong promise.

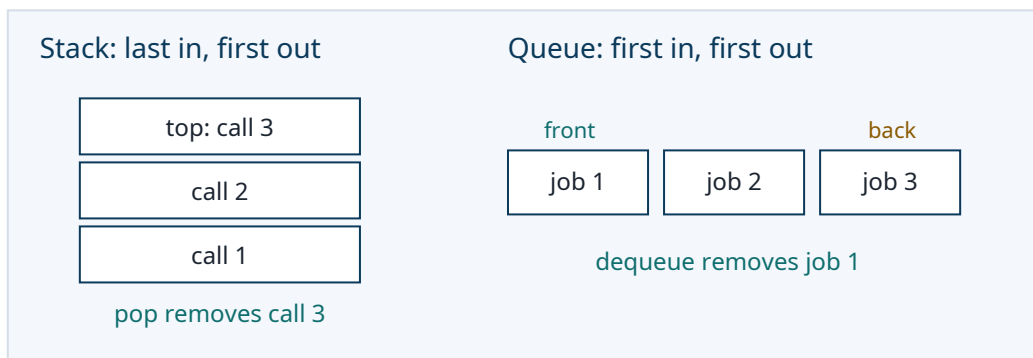


Figure 11. A stack removes the latest item. A queue removes the earliest item.

Trees

A **tree** is a set of **nodes** with one **root** and a parent-to-child relation, and no cycles. A **leaf** has no children. The **height** is the number of steps on the longest path from the root to a leaf; some books count nodes instead of edges, so say which you mean if you state a number. A **binary tree** gives each node at most two children, usually called left and right.

A **binary search tree** (BST) keeps an order. Every value in the left subtree is less than the node, and every value in the right subtree is greater. Duplicates need a stated policy; the trees in this lecture have none. The order is what makes a search able to ignore a whole subtree, the same idea as binary search.

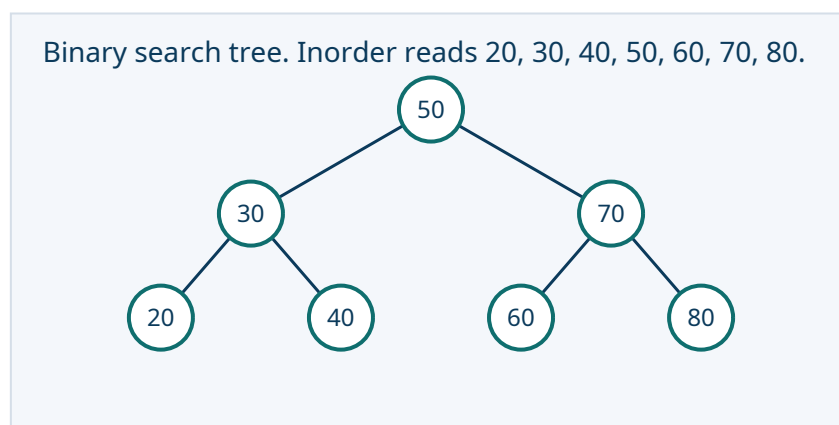


Figure 10. Left child smaller, right child larger. Inorder of this tree is sorted.

That tree is a BST: 30 and its children are less than 50; 70 and its children are greater; the same rule holds at 30 and at 70. Searching for 60 starts at 50, goes right because 60 is greater, and goes left from 70 because 60 is smaller. Two comparisons reach it. Searching for 25 goes left from 50 to 30, then left to 20, and stops: 25 is not there, and there is no further child to open.

Traversals

A **traversal** visits every node in a stated order. For a binary tree the three classical orders, applied at every node, are:

- **Preorder:** visit the node, then the left subtree, then the right subtree.
- **Inorder:** left subtree, then the node, then the right subtree.
- **Postorder:** left subtree, then the right subtree, then the node.

WORKED EXAMPLE — THREE ORDERS OF THE TREE ABOVE

Preorder (node, left, right): 50, 30, 20, 40, 70, 60, 80.

Inorder (left, node, right): 20, 30, 40, 50, 60, 70, 80. On a BST, inorder visits the values in sorted order. That is a check you should do.

Postorder (left, right, node): 20, 40, 30, 60, 80, 70, 50.

If your inorder of a BST is not sorted, the tree is not a BST or the traversal is wrong. Find out which.

Uses you should be able to name: inorder of a BST produces a sorted list; preorder copies a tree structure starting from the root; postorder is the natural order for deleting a tree, because the children go before the parent. A file system is a tree. An organisation chart is a tree if each person has one manager. A family tree with two parents is not a binary search tree and may not be a tree in the strict sense if you draw both parents; do not force it.

Breadth-first and depth-first walks

On a general tree or a small graph drawn as a hierarchy, two walks appear in questions. **Breadth-first search (BFS)** uses a queue. Visit the start, then its children left to right, then their children. **Depth-first search (DFS)** uses a stack, or recursion. Follow one child as far as it goes before the next sibling. If you visit a node when you first reach it, and you try children left to right, DFS matches preorder on a tree.

BFS from CEO: CEO, HR, IT, Staff, Dev, Accounts. The queue after visiting CEO holds HR then IT. DFS from CEO, children left to right, recording the node when first seen: CEO, HR, Staff, IT, Dev, Accounts. If a question says “search the Security department”, walk until the node named Security is visited and write the nodes in the order visited up to and including that node. Do not skip the nodes on the way. They are the trace.

Organisation tree used for the BFS and DFS traces.

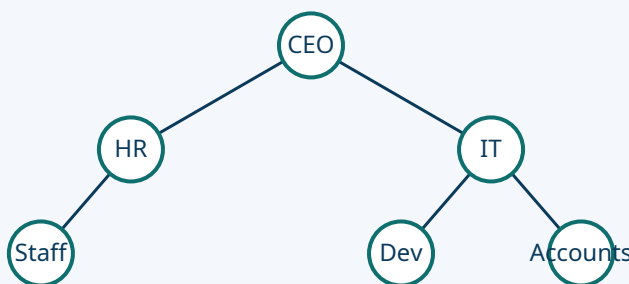


Figure 12. BFS visits level by level. DFS follows HR all the way to Staff before it opens IT.

COMMON MISTAKE

Inorder is not “top to bottom”. Preorder is not “left to right along the leaves”. Write the rule above the trace. If the question prints a tree, use that tree; a memorised 50-30-70 answer on a different diagram scores zero.

EXAM TIP

A three-mark traversal wants the order and nothing else. A five-mark question wants the order plus why that structure fits the problem. Keep the visit list on its own line so a marker can tick it.

Key terms

Term	Meaning in this lecture
Efficiency	How the work grows as the data grows.
Clarity	Whether another person can trace the steps.
Correctness	Whether every allowed input meets the specification.
Stack	Last-in, first-out. Push and pop.
Queue	First-in, first-out. Enqueue and dequeue.
Binary search tree	A binary tree ordered so left is smaller and right is larger.
Inorder	Left, node, right. Sorted on a BST.
Preorder	Node, left, right.
Postorder	Left, right, node.
BFS / DFS	Level by level with a queue, or down one branch with a stack.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Inorder of the BST rooted at 50 above. Write only the sequence.

MODEL ANSWER

20, 30, 40, 50, 60, 70, 80.

Why is a stack the wrong structure for a printer queue?

MODEL ANSWER

A stack prints the most recent job first. A printer queue promises the earliest job first, which is a queue.

Is the tree with root 10, left child 40, right child 30 a BST?

MODEL ANSWER

No. 40 is in the left subtree and is greater than 10, which breaks the BST rule.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Inorder traversal of a binary search tree produces

- A. the insertion order
- B. the values in sorted order
- C. only the leaves
- D. the values in reverse height order

2. BFS uses which structure to hold the waiting nodes?

- A. A stack
- B. A queue
- C. A primary key
- D. A hash

3. The main advantage of binary search over linear search on suitable data is that

- A. it works on unsorted data
- B. each comparison discards about half of the remaining items
- C. it never compares values
- D. it uses more memory and is therefore faster

4. Postorder visits a node

- A. before its children
- B. between its children
- C. after its children
- D. only if it is the root

Answer key with reasons

Q	Answer	Why
1	B	Left, then node, then right, walks a BST from smallest to largest.
2	B	The first node discovered is the first node expanded. That is a queue.
3	B	Halving is the advantage. It is invalid on unsorted data, so the first option is a common trap.
4	C	The rule is left subtree, right subtree, then the node.

Short questions

S1. Give one use of a stack and one use of a queue in computing, and the rule that makes each fit.

MODEL ANSWER

A call stack fits function calls because the last function called is the first to return. A printer queue fits because the first document submitted should be the first printed. Using the other structure would reverse the promise.

S2. Preorder and postorder of the tree rooted at 50 with the usual children 30, 70, 20, 40, 60, 80.

MODEL ANSWER

Preorder: 50, 30, 20, 40, 70, 60, 80. Postorder: 20, 40, 30, 60, 80, 70, 50.

S3. How do you judge efficiency without a formula, for linear versus binary search on a sorted list of a million names?

MODEL ANSWER

Linear search may examine every name. Binary search throws away about half the remaining names each time, so it still finishes after a small number of comparisons. Both can be correct. Binary search is the efficient one. Linear search may still be clearer to a beginner, which is a clarity comment, not an efficiency comment.

Long questions

L1. (5 marks) Insert 50, 30, 70, 20, 40, 60, 80 into an empty BST in that order. Give the inorder traversal and use it as a check. Then state how a search for 60 proceeds.

MODEL ANSWER

50 is the root. 30 goes left. 70 goes right. 20 goes left of 30. 40 goes right of 30. 60 goes left of 70. 80 goes right of 70. The drawing is the tree used throughout this lecture.

Inorder is 20, 30, 40, 50, 60, 70, 80, which is sorted, so the BST order is consistent.

Search for 60: compare with 50 and go right; compare with 70 and go left; compare with 60 and stop. The left subtree of 50 was never opened. That ignored subtree is the efficiency of the structure.

L2. (5 marks) A company stores departments as the tree CEO HR and IT; HR Staff; IT Dev and Accounts. A clerk must find Security, which is not in the tree, using DFS and then say what BFS would have visited first. Also recommend a structure if the real need is “who is the manager of this employee?” rather than a picture.

MODEL ANSWER

DFS from CEO, left child first, recording a node when first reached: CEO, HR, Staff, IT, Dev, Accounts. Security is never visited. The correct report is that the search ends after Accounts and Security is absent. Inventing a Security node is a wrong trace.

BFS visits CEO first, then HR and IT, then Staff, Dev, and Accounts. The first node is still CEO. BFS would discover a node near the root sooner than a node deep on the left, which is why it fits “shortest path in an unweighted tree”.

If the frequent question is the manager of a known employee, a tree drawing is a poor daily tool. A table of Employee(Id, Name, ManagerId) answers that lookup directly, and the tree can be drawn from it when a chart is wanted. Clarity and efficiency depend on the question, not on which picture looks more advanced.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 2 — Build a tree and walk it

- 1 Insert 8, 3, 10, 1, 6, 14, 4 into a BST on paper, in that order.
- 2 Write preorder, inorder, and postorder. Check that inorder is sorted.
- 3 Search for 6 and for 7. Record the nodes compared.
- 4 Write BFS and DFS visit orders from the root.

WHAT SUCCESS LOOKS LIKE

Inorder is 1, 3, 4, 6, 8, 10, 14. The search for 7 stops at a missing child and does not wander into 14.

Copy this summary

- Judge correctness, clarity, and efficiency separately. A fast wrong algorithm is still wrong.
- Stacks are last-in first-out. Queues are first-in first-out. Name a use that matches the rule.
- A BST puts smaller values on the left and larger values on the right.
- Preorder: node, left, right. Inorder: left, node, right. Postorder: left, right, node.
- Inorder of a BST is sorted. Use that as a check.
- BFS uses a queue and goes level by level. DFS goes down a branch first.
- Choose the structure for the question you actually ask, not for the most impressive name.

CLASS XII · UNIT 3

Lecture 3

Python: Objects, Files, and Databases

Textbook unit. Unit 3 — Programming Fundamentals

Curriculum domain. C. Programming fundamentals

Class time. about 40 periods

Class XI functions and lists are still the foundation. This lecture adds the reasons programming languages offer more than one style, then the Class XII toolkit: a small class, dictionaries, files, and a database you can normalise and query. Debugging moves from a single dry run to a unit test and a breakpoint.

Learning outcomes

- Compare object-oriented and functional styles at the level of purpose, benefits, and costs.
- Define a Python class with a constructor, attributes, and methods, and create two objects.
- Use lists, nested lists, and dictionaries, and say when a dictionary fits better.
- Read and write a text file, and count letters in a file.
- Explain normalisation to third normal form and write a filtered, ordered SQL query.
- Describe a unit test and use a print or a breakpoint to locate a fault.

Why there is more than one style

A **paradigm** is a style of organising a program so that a large problem stays understandable. **Imperative** code is the sequence, selection, and loops you already write. **Object-oriented programming** groups data and the operations on that data into objects, so a vehicle's mileage is updated by a vehicle, not by a loose function that might be handed the wrong variables. **Functional** programming emphasises functions that return results from inputs and avoid hidden changes to shared data, which makes some reasoning and testing easier. Python can host all three. This unit asks you to write a small object-oriented program and to say, in prose, what the functional emphasis would protect you from.

Style	What you gain	What it costs
Imperative	Direct steps, easy to trace	A long script becomes a tangle of shared variables
Object-oriented	Data and its operations stay together; pieces can be reused	A tiny task can be over-wrapped in classes
Functional	Functions are easier to test when they do not secretly change other data	Some problems read awkwardly if you ban every change

Table 12. A comparison you can reproduce in a short question. It is about organisation, not about which language is fashionable.

A class in Python

A **class** is the pattern. An **object** is one instance. **Attributes** are the data kept on the object. **Methods** are functions defined in the class; their first parameter is `self`, the object itself. `__init__` is the **constructor**: it runs when the object is created and usually stores the attributes. Outside the class you do not need to pass `self`. `Vehicle("Car", ...)` is enough.

```

Python
class Vehicle:
    def __init__(self, vehicle_type, brand, model, mileage):
        self.vehicle_type = vehicle_type
        self.brand = brand
        self.model = model
        self.mileage = mileage

    def update_mileage(self, new_mileage):
        self.mileage = new_mileage

    def display(self):
        print(
            self.vehicle_type,
            self.brand,
            self.model,
            self.mileage,
        )

car = Vehicle("Car", "Suzuki", "Alto", 12000)
van = Vehicle("Van", "Toyota", "Hiace", 54000)
car.update_mileage(12500)
car.display()
van.display()

```

The run prints Car Suzuki Alto 12500 and then Van Toyota Hiace 54000. The van is unchanged, which shows that each object has its own attributes. Encapsulation, at the level of this course, means those attributes are meant to be changed through methods such as `update_mileage` rather than by scattered assignments all over the program. Python will not stop you reaching in, but the method is the clear door.

Lists and dictionaries

A **nested list** is a list whose items are lists: one row per student, columns for id, name, and grade. A **dictionary** stores **values under keys**. Lookup by a key does not walk the whole collection the way a search of a list does. For a handful of rows you will not feel the difference. The design difference is already real: a dictionary says the id is the way you ask.

```

Python
rows = [
    [101, "Ali", 85],
    [102, "Babar", 90],
    [103, "Sana", 78],
]
by_id = {
    101: {"name": "Ali", "grade": 85},
    102: {"name": "Babar", "grade": 90},
    103: {"name": "Sana", "grade": 78},
}

print(by_id[103]["name"])
found = None
for row in rows:
    if row[0] == 103:
        found = row[1]
print(found)

```

Both prints are Sana. The dictionary reaches the key 103 directly. The nested list scans rows until the id matches. Use the list when the order of rows is the point, or when you will walk every row anyway. Use the dictionary when the question is “what do we know about this id?”. A list can be a value inside a dictionary, and a dictionary can sit inside a list. Do not nest them so deeply that you cannot say, in one sentence, what the outer collection means.

Files

Variables disappear when the program ends. A **file** keeps data on disk. `open` connects the program to a file. A mode of "w" starts a text file for writing and **replaces** existing contents. "a" appends. "r" reads. Always close the file, or use `with`, which closes it even if a later line fails. Disk input and output matters because results, logs, and datasets are larger than a screen and must still be there tomorrow.

```
Python
def count_letters(path):
    counts = {}
    with open(path, encoding="utf-8") as handle:
        text = handle.read().lower()
    for ch in text:
        if "a" <= ch <= "z":
            counts[ch] = counts.get(ch, 0) + 1
    return counts

with open("sample.txt", "w", encoding="utf-8") as handle:
    handle.write("Abba!")
print(count_letters("sample.txt"))
```

The file contains A, b, b, a, and an exclamation mark. After lowering, the counts are a: 2, b: 2. The exclamation mark is ignored because the test keeps only a to z. `dict.get(ch, 0)` returns 0 when the letter has not been seen, so the first time a letter appears the count becomes 1. This is the standard “occurrences of each letter” practical.

Normalisation

Normalisation organises tables so that a fact is stored once and update anomalies shrink. You should be able to reach **third normal form** on a small example and say what was wrong before.

- 1 **First normal form (1NF)**: each column holds one value, not a list. Repeating groups such as Subject1, Subject2 are removed.
- 2 **Second normal form (2NF)**: the table is in 1NF, and every non-key column depends on the whole key, not on part of it. This bites when the key is made of two columns.
- 3 **Third normal form (3NF)**: the table is in 2NF, and no non-key column depends on another non-key column. A fact about an instructor does not live in the course row if it depends only on the instructor.

WORKED EXAMPLE — FROM A WIDE ROW TO 3NF

Start with Result(StudentId, CourseId, StudentName, CourseName, InstructorId, InstructorOffice, Grade). The intended key is (StudentId, CourseId).

StudentName depends only on StudentId. CourseName and InstructorId depend only on CourseId. Those are partial dependencies, so the table is not in 2NF. Split:

Student(StudentId, StudentName) Course(CourseId, CourseName, InstructorId, InstructorOffice)
Result(StudentId, CourseId, Grade)

Result is now in 3NF: Grade depends on the whole key. Course is in 2NF but not in 3NF, because InstructorOffice depends on InstructorId, and InstructorId depends on CourseId. Split again:

Course(CourseId, CourseName, InstructorId) Instructor(InstructorId, InstructorOffice)

Changing an office is now one edit. Grade is still stored per student per course.

SQL you can write by hand

The queries in this course read a table, filter it, summarise it, and order it. Learn this pattern and you can answer the paper even if the lab uses a particular product.

SQL

```
SELECT name, salary
FROM employees
WHERE department = 'IT'
ORDER BY salary DESC;
```

```
SELECT AVG(salary) AS average_salary
FROM employees;
```

Suppose the rows are Hina, IT, 90000; Omar, Sales, 70000; and Sara, IT, 110000. The first query returns Sara then Hina, because of the descending salary order, and does not return Omar. The average of all three salaries is $(90000 + 70000 + 110000) / 3 = 90000$. `WHERE` filters before the average if you put it on that query. Without `WHERE`, Sales is included. Say which table the question printed before you compute.

A Python program can talk to a database through a library: open a connection, execute a statement, commit a change, close the connection. The paper may not ask for the connection syntax of a particular product. It will ask what the query returns and why the tables were split. If your lab uses SQLite, the same SQL above is enough.

Tests and breakpoints

A **unit test** calls one function or one method with a known input and checks the returned value or the new attribute. `celsius_to_fahrenheit(0)` should be 32. `update_mileage` on a fresh vehicle should leave the other vehicle alone. A test that requires you to type at the keyboard is a weaker test. **Breakpoints** pause a running program so you can see attributes at that line. A **watch** is the variable you have asked the debugger to show. Until the debugger is familiar, print the id and the grade just before the line you distrust, then delete that print.

COMMON MISTAKE

Opening a file with mode "w" to read it erases it first. Counting letters without `.lower()` treats A and a as different, which may or may not match the question. Storing InstructorOffice in every course row means one move of office must be edited many times, and one of those edits will be missed.

Key terms

Term	Meaning in this lecture
Paradigm	A style of organising a program.
Class / object	The pattern, and one instance of it.
Constructor	<code>__init__</code> , which stores the new object's attributes.
Dictionary	Values stored under keys, for lookup by key.
Nested list	A list of lists, often one row per record.
Unit test	A check that one function does one known job.
1NF, 2NF, 3NF	Atomic values; full dependence on the key; no dependence on another non-key column.
WHERE / ORDER BY	The filter and the sort of an SQL query.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

What does the sample Vehicle program print for the car after update_mileage(12500)?

MODEL ANSWER

Car Suzuki Alto 12500. The van remains Van Toyota Hiace 54000.

Why is InstructorOffice removed from Course on the way to 3NF?

MODEL ANSWER

It depends on InstructorId, not on CourseId. InstructorId is not the key of Course. A non-key column depending on another non-key column violates 3NF.

What does count_letters return for a file whose text is Abba! ?

MODEL ANSWER

A dictionary with a: 2 and b: 2. The exclamation mark is ignored and the letters are lowered before counting.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. A primary benefit of object-oriented programming, at this level, is that

- A. it forbids all functions
- B. data and the operations on it can be kept and reused together
- C. it removes the need for variables
- D. it is always the fastest possible style

2. Mode "w" in open

- A. appends safely
- B. replaces the file's contents for writing
- C. reads only
- D. normalises to 3NF

3. AVG(salary) without a WHERE clause

- A. averages only the IT rows
- B. averages every row of the table
- C. sorts the table
- D. deletes the Sales rows

4. A unit test is most useful when the function

- A. returns a value you can check
- B. only clears the screen
- C. has no name
- D. must be retyped by the user on every run

Answer key with reasons

Q	Answer	Why
1	B	The point of a class is to keep related data and behaviour together. The other options are false.
2	B	Write mode truncates an existing file. Append is "a". Read is "r".
3	B	The filter is what restricts the average. No filter means the whole table.

Q	Answer	Why
4	A	The test compares a known input with an expected result. A return value makes that comparison direct.

Short questions

S1. Write a Python dictionary for the three students Ali 85, Babar 90, and Sana 78, keyed by 101, 102, and 103. How do you print Sana's grade?

MODEL ANSWER

`by_id = {101: {"name": "Ali", "grade": 85}, 102: {"name": "Babar", "grade": 90}, 103: {"name": "Sana", "grade": 78}}`. Print `by_id[103]["grade"]`, which is 78. The outer key is the id. The inner keys are the field names.

S2. State one partial dependency and one transitive dependency in the wide Result table from this lecture.

MODEL ANSWER

StudentName depends only on StudentId, which is part of the key (StudentId, CourseId). That is partial. After Course is split out, InstructorOffice depends on InstructorId rather than on CourseId. That is transitive, and it is why Instructor becomes its own table.

S3. What should a unit test for update_mileage check?

MODEL ANSWER

Create a vehicle with mileage 100, call `update_mileage(150)`, and check that the attribute is 150. Create a second vehicle and check that its mileage did not change. A test that only prints the first vehicle can hide the second fault.

Long questions

L1. (5 marks) Design a Vehicle class in Python with type, brand, model, and mileage. Include a constructor, a method to update mileage, and a method to display the vehicle. Show two objects and one update.

MODEL ANSWER

The class in this lecture is the answer. The constructor stores the four attributes on self. `update_mileage` replaces `self.mileage`. `display` prints the four attributes. `car = Vehicle("Car", "Suzuki", "Alto", 12000)` and `van = Vehicle("Van", "Toyota", "Hiace", 54000)` create two independent objects. `car.update_mileage(12500)` changes only the car. The printed lines are Car Suzuki Alto 12500 and Van Toyota Hiace 54000.

A functional style would instead return a new description without changing an object. The object-oriented version fits here because the mileage is a lasting fact about one vehicle and several operations share those attributes.

L2. (5 marks) Normalise Result(StudentId, CourseId, StudentName, CourseName, InstructorId, InstructorOffice, Grade), with key (StudentId, CourseId), to 3NF. Then write a query idea, in SQL or in clear words, for the average grade of one course.

MODEL ANSWER

Student(StudentId, StudentName). Instructor(InstructorId, InstructorOffice). Course(CourseId, CourseName, InstructorId). Result(StudentId, CourseId, Grade). StudentName was partial on StudentId. Course facts were partial on CourseId. InstructorOffice was transitive through InstructorId and therefore left the Course table.

```
SELECT AVG(Grade) AS average_grade FROM Result WHERE CourseId = 'CS12';
```

The WHERE must be present or the average mixes every course. The student name is not in Result anymore, which is the point of the split: you join Student if you need the name, and you do not repeat the name on every grade.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 3 — Class, file, and a query

- 1 Implement Vehicle and print the two lines from the worked example.
- 2 Write sample.txt containing Abba! and print the letter counts. Add a capital letter and confirm it merges after lowercasing.
- 3 Create the 3NF tables in SQLite or on paper. Insert two students, two instructors, two courses, and three results.
- 4 Run the average-grade query for one course and check it by hand.
- 5 Add one assert, or a small test function, that update_mileage does not change the other object.

WHAT SUCCESS LOOKS LIKE

The car prints 12500, the letter count shows a and b, and the SQL average matches the hand average of that course only.

Copy this summary

- Paradigms organise programs. Objects keep data with behaviour. Functions are easier to test when they do not change hidden state.
- A class has a constructor, attributes, and methods. Each object keeps its own attributes.
- A dictionary answers “this key”. A nested list keeps rows in order and is scanned.
- Use with to open a file. Mode w replaces the file. Lowercase before counting letters if case should not matter.
- 1NF removes repeating groups. 2NF removes partial dependencies. 3NF removes transitive dependencies.
- WHERE filters. ORDER BY sorts. AVG without WHERE covers the whole table.
- A unit test checks one known result. A breakpoint shows the variables in the middle of a run.

CLASS XII · UNIT 4

Lecture 4

Data, Models, and Hypothesis Tests

Textbook unit. Unit 4 — Data and Analysis

Curriculum domain. D. Data and analysis

Class time. about 25 periods

Class XI fitted a straight line and refused to call it a cause. Class XII asks how a model learns from examples, how you tell a good prediction from a flattering one, and how a hypothesis test keeps a lucky pattern from being announced as a discovery. You also tell the story of a dataset with a chart or a query, not with a single dramatic number.

Learning outcomes

- Contrast a rule-based method with machine learning, and describe features, a train-test split, and a metric.
- Compute precision from a confusion matrix and say what the number does not mean.
- State a null and an alternative hypothesis, and use a p-value against a significance level.
- Choose a visualisation or an SQL summary that answers a stated question, and limit the claim.

Rules and learning

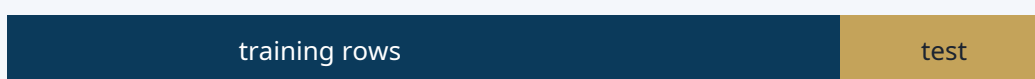
A **rule-based** method follows instructions a person wrote: if attendance is below 75 percent, do not allow the exam; if the mark is at least 40, record a pass. The rule does not change when new rows arrive, unless a person edits it. **Machine learning** builds a model by adjusting it on examples. The model is then used to predict a label for a new row. It is useful when the rule is hard to write by hand, such as “does this photograph look like a cotton leaf with a particular damage?”, and dangerous when people treat the prediction as a cause or as a verdict that nobody may appeal.

A **feature** is an input the model is allowed to see, such as hours of practice or the number of past quizzes attempted. A **label** is the outcome on the training rows, such as pass or fail, or the quiz mark. **Feature engineering** is the work of turning raw records into those inputs: three monthly tests become one average, a date of birth becomes an age, a free-text comment is left out because the model was not given a careful way to read it. A bad feature list produces a confident wrong model. Leaving out a feature that simply copies the answer (for example, including “passed” as a feature when predicting “passed”) is a cheat, not engineering.

Train, test, and the metrics

If you measure a model on the same rows it learned from, it can memorise them and look brilliant. A **train-test split** holds some rows back. The model learns only on the training rows. The test rows estimate how it behaves on data it has not memorised. A common classroom split is most of the rows for training and the rest for testing, chosen so that both parts still contain the kinds of cases you care about. Splitting one student’s repeated rows into both sides can leak the answer. Say how you split.

Train on most rows. Measure on rows the model has not seen.



A score on the training rows can be memorisation. The test rows are the check.

Figure 13. Keep the test rows out of training, or the score flatters the model.

A **confusion matrix** counts the test predictions against the truth. For a yes/no prediction:

	Predicted positive	Predicted negative
Actual positive	True positive (TP)	False negative (FN)
Actual negative	False positive (FP)	True negative (TN)

Table 13. Names of the four counts. Positive means the event you are trying to detect.

Accuracy is $(TP + TN) / \text{all rows}$. **Precision** is $TP / (TP + FP)$: of the rows you called positive, how many were. **Recall** is $TP / (TP + FN)$: of the truly positive rows, how many you caught. A model can have high accuracy and still be useless if the event is rare and it never predicts the rare event. Choose the metric the decision needs. A scholarship screen that must not miss a qualifying student cares about recall. A screen that triggers an expensive interview cares about precision.

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

WORKED EXAMPLE — PRECISION

Predicted positive and actually positive: 90. Predicted negative and actually positive: 10. Predicted positive and actually negative: 30. Predicted negative and actually negative: 70.

TP = 90, FN = 10, FP = 30, TN = 70. There are 200 rows.

Precision = $90 / (90 + 30) = 90 / 120 = 0.75$, which is 75 percent. Recall = $90 / (90 + 10) = 90$ percent.

Accuracy = $(90 + 70) / 200 = 80$ percent.

The 75 percent does not mean “75 percent of students passed”. It means three quarters of the positive predictions were right.

A **hyperparameter** is a setting you choose rather than a value the model learns from the labels, such as how many groups to look for, or how flexible the model is allowed to be. Trying many settings and keeping the one that looks best on the *test* set makes the test set part of the training. The honest report says which data the choice was based on. You are not required to tune a large model in this course. You are required to know that the setting is a choice and that the metric must match the decision.

PREDICTION IS NOT A CAUSE

A model that predicts quiz marks from practice hours has found an association in the rows it was given. It has not proved that changing the hours will change the marks. Class XI’s warning still stands. Do not write “the model shows that practice causes success”. Write “the model predicts, on similar rows, with this error rate”.

Hypothesis testing

A **hypothesis** is a claim you can challenge with data. The **null hypothesis (H0)** is the default of no effect or no difference: the new study technique has the same mean score as the old one; the coin is fair. The **alternative hypothesis (H1)** is the challenge: the means differ, or the new mean is higher, or the coin is not fair. Write both before you look at a dramatic number. Changing the hypothesis after you see the data is a different, weaker story, and you should admit it if you did it.

A **p-value** is the probability of seeing a result at least as extreme as yours **if the null hypothesis is true**. It is not the probability that the null is true. A **significance level**, written alpha, is the threshold you chose in advance, often 0.05. If the p-value is smaller than alpha, you **reject the null**. If it is not, you **do not reject the null**. “Do not reject” is not the same sentence as “the null is true”. It means this dataset did not supply strong evidence against it.

WORKED EXAMPLE — A COIN, SMALL ENOUGH TO COUNT

H0: the coin is fair, so the chance of heads on a toss is 0.5. H1: the chance of heads is greater than 0.5. Alpha = 0.05. You plan to toss 10 times.

You observe 9 heads. Under a fair coin, the number of ways to get 9 or 10 heads is $10 + 1 = 11$, and there are $2^{10} = 1024$ equally likely sequences. The one-sided p-value is $11/1024$, about 0.011.

0.011 is less than 0.05, so you reject the null at this level. You have not proved the coin is biased. You have said that nine or ten heads would be unusual for a fair coin, and you saw that kind of result. Ten tosses are a classroom count, not a pub trick to accuse a person. Say the sample size in the same breath as the decision.

A study-technique question uses the same shape and usually cannot be computed by hand from a binomial. You still write H0 and H1, you name alpha, you say you would compare the p-value with alpha, and you tie the conclusion to a chart: the two groups' score distributions, not a slogan. If a question gives you the p-value, use it. Do not invent one.

Telling the story of the data

Data storytelling is a sequence: the question, the source of the rows, the picture, and the sentence you are willing to defend. Several views beat one view. A mean can rise while the median stays still because of a few high scores. A bar chart of averages and a note of the sample size answer different doubts. **Descriptive statistics** — mean, median, spread — describe the rows you have. They are not yet a hypothesis test.

SQL belongs here as well as in programming. `SELECT Subject, AVG(Mark) FROM Result GROUP BY Subject` is a summary a chart can use. A Python plot or a spreadsheet chart of that summary is the visualisation. Criticise a published chart the way you did in Class XI: missing axis, a truncated scale, a pie of things that are not parts of a whole, a claim about a population the sample does not represent.

EXAM TIP

If the question gives a matrix, compute the ratio it names and state the formula in symbols or words before the arithmetic. If it gives a study, write H0 and H1 in the language of that study. A generic definition copied beside an unused scenario loses the application marks.

Key terms

Term	Meaning in this lecture
Rule-based method	Fixed instructions written by a person.
Machine learning	A model adjusted on examples and then used to predict.
Feature / label	An input column, and the outcome the model is asked to predict.
Train-test split	Learn on some rows; measure on rows that were held back.
Precision	$TP / (TP + FP)$. How many positive predictions were right.
Recall	$TP / (TP + FN)$. How many real positives were caught.
Null hypothesis	The claim of no effect or no difference, written before the test.
P-value	How surprising the data are if the null is true. Not the probability the null is true.

Term	Meaning in this lecture
Significance level	The threshold, alpha, below which you reject the null.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Precision is 75 percent in the worked matrix. What is the matching fraction?

MODEL ANSWER

$90 / (90 + 30) = 90/120 = 0.75$. The 30 are the false positives.

Why hold back test rows?

MODEL ANSWER

A model can memorise the training rows. The held-back rows estimate how it behaves on data it was not shown.

A p-value of 0.20 and alpha of 0.05. What do you decide?

MODEL ANSWER

Do not reject the null. 0.20 is not smaller than 0.05. This is not proof that the null is true.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. The train-test split exists so that

- A. the model is scored on rows it learned from
- B. the model is scored on rows it did not learn from
- C. every row is used twice
- D. the null hypothesis becomes true

2. Alpha is

- A. the probability you accept a false null
- B. the threshold for rejecting the null when it is true, chosen in advance
- C. the sample size
- D. the mean of the sample

3. Feature engineering means

- A. hiding the test set inside the training set
- B. turning raw records into the inputs a model will use
- C. deleting the metric
- D. replacing SQL

4. A high accuracy on a rare event can still be a weak model if

- A. it never predicts the rare event
- B. it uses a confusion matrix
- C. precision is defined
- D. the chart has a title

Answer key with reasons

Q	Answer	Why
1	B	Held-back rows estimate performance beyond memorisation.
2	B	Alpha is the significance level: the false-rejection risk you decided to tolerate if the null is true.
3	B	Features are the prepared inputs. Including the answer itself as a feature is a fault, not a goal.
4	A	Always saying “no” can be accurate when positives are rare, and it catches none of them. Recall exposes that.

Short questions

S1. Define precision and compute it when TP = 90 and FP = 30.

MODEL ANSWER

Precision is the share of positive predictions that were actually positive, $TP / (TP + FP)$. Here $90 / 120 = 0.75$, or 75 percent.

S2. Write H0 and H1 for a test of a new study technique against the usual one, two-sided.

MODEL ANSWER

H0: the mean score with the new technique equals the mean score with the usual technique. H1: the two means differ. A one-sided H1 would instead say the new mean is higher. The question asked for two-sided, so “differ” is the right alternative.

S3. Give one SQL summary and one chart that could present average marks by subject.

MODEL ANSWER

SELECT Subject, AVG(Mark) AS average_mark FROM Result GROUP BY Subject. A bar chart of those averages, axis from zero, with the number of students per subject written somewhere, presents the comparison. A pie chart of averages is the wrong picture.

Long questions

L1. (5 marks) A model predicts whether a student will score 80 or above. On the test set, TP = 90, FN = 10, FP = 30, TN = 70. Compute precision, recall, and accuracy. Say which metric you would watch if the cost of a false alarm (extra tutoring offered to someone who would have scored well anyway) is the main worry, and state one thing the model has not proved.

MODEL ANSWER

Precision = $90 / (90 + 30) = 75$ percent. Recall = $90 / (90 + 10) = 90$ percent. Accuracy = $(90 + 70) / 200 = 80$ percent.

A false alarm is a false positive. Precision is the metric that falls when false positives grow, so it matches that worry. Recall would matter more if missing a student who needed support were the expensive mistake.

The model has not proved that any particular habit causes a distinction. It has classified held-back rows with these error counts, and only for data like the data it was given.

L2. (5 marks) A teacher tosses a coin 10 times to demonstrate a test and gets 9 heads. Take H0 as “fair coin”, H1 as “heads is more likely than tails”, and alpha as 0.05. Use p 0.011. Decide, and explain the decision in a way that would also fit a study-technique experiment whose p-value you were given.

MODEL ANSWER

The p-value is about 0.011, which is smaller than 0.05, so reject H_0 at the 5 percent level. Under a fair coin, 9 or 10 heads in 10 tosses is uncommon (11 sequences out of 1024). Rejecting H_0 does not certify the coin, and $n = 10$ should be stated.

The same sentences fit a study technique. Write H_0 as equal means and H_1 as the direction you chose. If the given p-value is below alpha, reject H_0 and show the two score distributions so the reader sees the difference the test is about. If the p-value is above alpha, do not reject H_0 , and do not rewrite H_1 after peeking. In both stories the p-value is the surprise of the data assuming the null, not the probability that the null is true.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 4 — A metric and a hypothesis write-up

- 1 Copy the confusion matrix from the worked example and compute precision, recall, and accuracy by hand and in a few lines of Python.
- 2 Write H_0 , H_1 , alpha, and the decision for the coin example in your journal.
- 3 Using any small table of marks, produce an average by subject in SQL or in a spreadsheet and chart it.
- 4 Under the chart, write one causal sentence you refuse to make.

WHAT SUCCESS LOOKS LIKE

The three percentages are 75, 90, and 80. The chart has a title and a sample size. The refused sentence is written down, not only implied.

Copy this summary

- Rules are written by people. Machine learning adjusts a model on examples. Neither is a proof of cause.
- Features are inputs. Labels are known outcomes. Do not feed the model the answer as a feature.
- Train on some rows and measure on held-back rows.
- Precision = $TP/(TP+FP)$. Recall = $TP/(TP+FN)$. Accuracy can hide a rare event.
- H_0 is no difference. A p-value is surprise assuming H_0 . Compare it with alpha. Do not reject is not proof.
- Tell the story with more than one view: a summary query, a chart, and a limit on the claim.

CLASS XII · UNIT 5

Lecture 5

Applications, Privacy, and Safe Collaboration

Textbook unit. Unit 5 — Applications and Impacts of Computing

Curriculum domain. E, F, and G. Applications, impacts, and digital research

Class time. about 40 periods

Class XI introduced IoT, cloud, blockchain, bias, and the digital divide. Class XII asks you to design an application for a Pakistani setting, to explain deep learning without advertising it, and to handle the conflicts that appear when data is shared. It also asks how you work safely with other people online, and how you publish a finding with digital tools.

Learning outcomes

- Propose an application for Pakistan that uses IoT, cloud computing, or blockchain, including a risk.
- Explain deep learning as layered neural networks that learn features, and name a use and a limit.
- Describe a data-sharing conflict and a policy compromise.
- Name common attacks by their impact and match them to 2FA, biometrics, and encryption in transit.
- Collaborate more safely, and publish a small research artifact with an advanced search behind it.

An application for a real setting

A design answer has a user, a job to be done, the data, which of IoT, cloud, or blockchain is actually needed, and a risk. Adding all three technologies to sound modern is a weak answer. A soil sensor does not need a blockchain. A degree check might use a hash on a shared ledger and does not need a moisture sensor.

WORKED EXAMPLE — FLOOD MARKS ALONG A RIVER

User: a district officer and the villages upstream of a known low road.

Job: know when a water level is rising faster than usual, early enough to close the road.

Data: a water-level reading every few minutes, the sensor's id, and the time. Not the names of residents.

Technology: IoT sensors send readings to a cloud dashboard. The cloud is justified because more than one office must see the same history, and a local memory card in a drowned box is a bad archive.

Blockchain is not justified.

Risk: a failed radio looks the same as a calm river if the dashboard does not show “sensor silent”. The design must alarm on silence, not only on high water.

Deep learning

A **neural network** is a model made of many simple units in layers. Early layers transform the input. Later layers produce a prediction. **Deep learning** means a neural network with many layers, trained on a large number of examples. Its practical claim is that it can learn **features** from raw input — pixels, sound samples — instead of waiting for a person to invent every feature. That is why it shows up in image recognition, speech, and translation.

The limits belong in the same paragraph. It needs a lot of examples and a lot of computing. If those examples are skewed, the errors are skewed. It can be hard to explain which part of a photograph drove a decision. A crop-disease photo tool may help a farmer decide to ask an expert. It is not itself the expert, and a Class XII answer should not claim a medical or agricultural diagnosis.

	Traditional machine learning	Deep learning
Features	Often designed by a person	Often learned from raw data
Data and compute	Can work on smaller sets	Usually wants much more of both
Explanation	Sometimes easier to point at a feature	Often harder to explain a single decision
A fitting job	Predict a mark from a few numbers	Classify a photograph or a sound

Table 14. Traditional machine learning versus deep learning, for a short comparison question.

Sharing data and keeping a boundary

Data sharing lets another party do something useful: a hospital shares case counts with a health department, a university confirms to an employer that a degree is real. **Privacy** says the person the data is about still has a boundary. The two goals conflict as soon as the useful share reveals more than the job needs. There is no design in which everybody gets everything and nobody is exposed. A policy is a compromise you can defend.

Useful compromises are specific. Share counts, not names. Share a yes/no check of a degree, not the transcript. Keep the detailed rows with the organisation that collected them, and let outsiders ask a narrow question. Write down who may receive the data and for how long. Give the person a way to ask what is held. When stakeholders disagree — the company wants an advertisement profile, the student wants to be left alone, the college wants alumni news — the answer should say whose interest you would not sacrifice and why. “Balance” without a decision is not a policy.

Attacks and the controls that match them

You should recognise the following by the harm they cause, and refuse to confuse them. **Phishing** is a message that imitates a trusted party in order to collect a secret or a payment. **Malware** is software that harms a device or the data; **ransomware** is malware that locks files and demands payment; **spyware** is malware that reports what you do. A **virus** spreads by attaching to other files people open. A **DDoS** attack floods a service until genuine users cannot reach it. These are different jobs. A strong password does not stop a flood of a public website, and a backup does not by itself stop a phishing page.

Control	What it improves	The limit
2FA	A stolen password is not enough to log in	A one-time code can itself be phished if the user is rushed
Biometric check	The person is present, and they need not remember a long secret	A fingerprint database leak cannot be “reset” like a password; some people cannot use the sensor
Encryption in transit	Someone on the path sees ciphertext, not the message. HTTPS is the ordinary web example	It does not make a fraudulent website honest. The lock icon is not a review of the business
Backup	Ransomware or a failed disk does not end the only copy	A backup that shares the same password and the same infection is a weak backup
Least privilege	A stolen clerk account cannot do an administrator’s job	It does nothing if everyone shares one administrator login

Table 15. Controls to pair with threats. Describe the control; do not invent a system to carry the attack out.

Safe collaboration uses the same controls. A shared folder has named people, not a link that anyone on the Internet can edit. A group project does not live in one student’s personal account with the password handed around. You agree what must not be uploaded: another person’s identity card, a full set of marks with names, a password “so you can submit for me”. Identity theft becomes easier when those documents are scattered across chats. The practical habit is to collect less and to share the document, not the login.

Publishing a small research artifact

Class XII's digital-literacy outcome is an **artifact**: a short report, a slide deck, or a notebook that answers a question and shows the evidence. The search behind it should use operators from Class XI, and the page should be judged, not merely found. The artifact itself has a question, a method, a chart or a quoted official figure with a date, a conclusion, and a limit. Digital tools help because the chart, the source link, and the revision history are visible. They do not replace the reading.

WORKED EXAMPLE — A ONE-PAGE ARTIFACT

Question: what does our college actually collect before a student can use the lab network?

Method: read the college's own notice (secondary, official) and interview the lab in-charge with five questions (primary). Search used: the college name plus `site:.edu.pk` and the word "laboratory".

Finding: the notice asks for a roll number and a parent phone number. The in-charge also keeps a photocopy of the identity card, which the notice does not mention.

Conclusion, limited: in this college the practice is wider than the notice. Recommendation: stop the photocopy or add it to the notice and say who may see it. This is a policy suggestion for one college, not a law you have drafted for the country.

EQUITY

A design that only works in English, only on a new phone, or only with a camera on, shares its benefits with the students who already had the most access. When you propose an application, say how a student on a shared basic phone still gets the essential message, even if that message is an SMS rather than a dashboard.

Key terms

Term	Meaning in this lecture
Deep learning	Neural networks with many layers, trained to learn features from raw data.
Neural network	A layered model of simple units that produces a prediction.
Data sharing	Giving another party access to data for a stated job.
Privacy	The boundary around a person the data is about.
Phishing	A fake message aimed at stealing a secret or a payment.
Ransomware	Malware that locks files and demands payment.
2FA	A login that needs a second, different proof.
Artifact	A digital product that states a finding and the evidence.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Why is blockchain a mismatch for the flood-sensor example?

MODEL ANSWER

The problem is timely readings and a shared dashboard, which a cloud store already provides. A shared ledger does not detect a silent sensor and adds a design the user did not need.

What does encryption in transit fail to guarantee?

MODEL ANSWER

It hides the contents on the path. It does not prove the site is honest. A phishing site can also show a lock.

Name one equity feature for a parent who has a basic phone.

MODEL ANSWER

Send the essential alert as an SMS, not only as a page on a dashboard that the phone cannot open.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. Deep learning differs from a small traditional model mainly because it

- A. never uses data
- B. can learn features from raw inputs such as pixels, using many layers
- C. is always easy to explain
- D. does not make errors

2. A DDoS attack primarily harms

- A. availability
- B. the grammar of SQL
- C. inorder traversal
- D. the mean of a sample

3. The better share of degree data with an employer is

- A. the full school file of every student
- B. a check that this named degree was awarded
- C. the student's home address and marks in every class
- D. a public copy of the identity card

4. 2FA helps most against

- A. a flood of traffic on a public site
- B. a stolen password used from elsewhere
- C. a lost backup disk that was never encrypted
- D. a chart with no axis

Answer key with reasons

Q	Answer	Why
1	B	Many layers and learned features are the distinction. Explanation and perfection are not promised.
2	A	The service is flooded so genuine users cannot use it. That is an availability failure.
3	B	The job is verification. The other options share far more than the job needs.
4	B	The second factor blocks a password that is no longer secret. It does not stop a traffic flood.

Short questions

S1. Give one Pakistani IoT idea and the risk you would mention in the same answer.

MODEL ANSWER

Moisture sensors in a field send readings to a cloud store, and the farmer gets an SMS when the soil stays dry. The risk is a dead battery or a missing signal looking like “no problem” unless the system alerts on silence. Names of the farmer's family are not required.

S2. Contrast phishing with ransomware in one sentence each, and name one control for each.

MODEL ANSWER

Phishing tricks a person into handing over a secret; a control is to open the service from a bookmark and to use 2FA so a captured password is incomplete. Ransomware locks files; a control is a backup that is not continuously open to the infected machine, plus not opening unexpected attachments. Paying is not a control you should describe as reliable.

S3. What four parts should a one-page research artifact have?**MODEL ANSWER**

A question, a method that says whether the data is primary or secondary, a result the reader can check, and a conclusion that does not outrun the sample. A date on the sources belongs with the method.

Long questions**L1. (5 marks) Design a degree-verification idea using cloud computing or blockchain. Explain the choice, the privacy compromise, and one stakeholder who may dislike it.****MODEL ANSWER**

The user is an employer who wants to know whether a named person was awarded a named degree. The university keeps the full record. What is shared is a check: yes or no, plus the year and the title of the award. A hash of that record, or a signature the university can verify, can be looked up without copying the transcript. A blockchain fits only if several universities and employers need a ledger none of them alone controls. A single university can do the same job with a cloud service and a signed answer, which is simpler. Choose one and say why.

Privacy: the employer does not receive address, grades, or identity-card scans. The student should be able to see who asked. A stakeholder who may dislike it is a vendor of paper certificates, or an employer who wanted the full transcript for other, less declared uses. The policy refuses that wider share.

L2. (5 marks) Your project group works in a shared online folder and on a chat. Describe three unsafe habits and the safer practice for each. Then say how you would present the group's findings so a marker can check them.**MODEL ANSWER**

Sharing one login is unsafe because you cannot tell who changed a file and you cannot remove one person later. Invite named accounts instead. Posting a classmate's identity card "for the report" is unsafe because that image is enough for identity theft. Collect roll numbers only if the question needs them, and do not put them in the chat. A public editing link is unsafe because strangers, not only the group, can change the work. Restrict the folder to the group and the teacher.

The artifact is a short report or a few slides: the question, the method, one chart with n, the sources with dates, and a limit. The folder's history shows who wrote what. The chat is not the report.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 5 — A design page and a safer folder

- 1 Write a one-page design: user, job, data, one technology, one risk, one equity note.
- 2 Write a privacy rule for that design in two sentences: what is shared, what is not.
- 3 Create a shared folder for your pair with named access. Record one unsafe alternative you refused.
- 4 Add a five-line artifact that cites one source you found with a search operator.

WHAT SUCCESS LOOKS LIKE

The design names a technology it refused, as well as one it used. The folder is not shared by a single password.

Copy this summary

- Use IoT, cloud, or blockchain only when the job needs it. Name the risk and the user.
- Deep learning learns features in many layers from raw data. It wants data and compute, and it can copy bias.
- Share the minimum that does the job. A policy names who gets what, not a vague balance.
- Phishing, ransomware, and DDoS are different harms. Match 2FA, backups, and service-level defences correctly.
- Encryption in transit hides the message on the path. It does not make the sender honest.
- Collaborate with named access. Do not pass logins or identity documents through a chat.
- An artifact has a question, a method, a checkable result, and a limited conclusion.

CLASS XII · UNIT 6

Lecture 6

Entrepreneurship in the Digital Age

Textbook unit. Unit 6 — Entrepreneurship in the Digital Age

Curriculum domain. H. Entrepreneurship

Class time. about 10 periods

A business idea is a guess about a person and a problem. This lecture is about testing the guess cheaply: choose a beachhead market, name the riskiest assumption, and build a minimum viable product that can prove you wrong. Class XI's paper prototype showed that a design could be used. The MVP asks whether the business idea survives contact with that user.

Learning outcomes

- Distinguish a prototype from a minimum viable product.
- Define a beachhead market narrowly enough to test.
- Identify the riskiest assumption and design a test that could fail.
- Describe an ethical digital offer a student could actually run, and name a career path the work points toward.

Prototype and MVP

A **prototype** is a learning model. It can be paper. Its question is “can a person understand and use this?” A **minimum viable product (MVP)** is the smallest real offer that lets a user complete the core job, so you can measure behaviour. Its question is “will the people in this market actually take this offer?” A slide that pretends to be an app is still a prototype. A single paid workshop, a working signup that delivers one chapter you wrote, or a weekly SMS that a farmer pays to receive, can be an MVP if a stranger to the design can finish the job.

MINIMUM VIABLE PRODUCT

The smallest version of a product that is real enough for the target user to complete the main job, built so that the riskiest assumption can be tested. It is not a buggy version of every feature you hope to add later.

The beachhead

A **beachhead market** is the first small group you will win, not the whole country you will mention in a speech. “Students” is not a market. “Class XI computer-science students in three colleges in one town, who already have a phone and who told a teacher they want extra practice on trees” is a beachhead. You can reach them, you can hear them, and a failure is visible. If the beachhead does not care, a larger market will not repair the idea.

The riskiest assumption

An idea is a pile of assumptions: the problem is real, this person feels it, they will try your solution, they will pay this price, you can deliver, and a rule or a parent or a college will allow it. The **riskiest assumption** is the one that, if false, kills the idea, and that you are least sure about. Often it is “they will pay”, or “they will come back next week”, not “we can draw a logo”.

A real test can fail. If you only ask friends “would you like this?” they will say yes. A stronger test asks for a behaviour: a small payment, a second visit, a reply to an SMS, a form filled without you standing over the desk. Decide before the test what result would make you change or stop. Changing the price after everyone refused, and then announcing success, is not a test.

WORKED EXAMPLE — PRACTICE CARDS FOR ONE BEACHHEAD

Beachhead: Class XII CS students in your own college who have a phone.

Offer: each week, one original practice card on a single topic (this week, inorder traversal), plus a short answer they can check. Delivered in a chat they already use.

Riskiest assumption: at least 15 students will complete the card two weeks in a row without the teacher making it compulsory. The polite assumption “they will say it is useful” is weaker, because it cannot fail in public.

MVP: you write two original cards, not a full app, not a dump of a textbook. You send them. You count completions. If fewer than the number you set in advance come back, you stop or you change the topic. You do not paste pages from the Sindh Textbook Board book or from another author’s notes and call it a product.

What you may not sell

A digital business still has rules. You do not sell copied books, copied notes, or a paper that a student will submit as their own work. You do not collect identity cards “for registration” and leave them in a chat. You do not invent a result, a ranking, or a customer count. You do not spam people who did not ask. An MVP that grows by those methods has not validated a business. It has tested how long a nuisance can continue. The same privacy and usability lessons from the earlier units apply to something you run yourself: collect less, label the action honestly, and let the user stop.

A test plan you can write in an examination

- 1 Name the beachhead in one sentence a stranger could use to find those people.
- 2 State the core job in one sentence, from the user’s side.
- 3 State the riskiest assumption.
- 4 Describe the MVP as the smallest thing that tests that assumption.
- 5 State the number or the behaviour that will count as failure.
- 6 Say what you will do if it fails: change the offer, change the beachhead, or stop.

WORKED EXAMPLE — FAILURE WRITTEN IN ADVANCE

Assumption: shopkeepers on one street will reply to a weekly SMS that lists items they asked you to watch in nearby markets.

MVP: ten shopkeepers, one week, messages you type yourself. No app.

Failure: fewer than six reply with a real question or a thank-you that includes a new item. A family member replying does not count. If it fails, you do not build an app. You either visit the shops and change what the message contains, or you drop the idea.

Careers the subject is pointing at

The units in these two years line up with real work. Building and testing software is software development and quality work. Databases and SQL lead toward data and information-system roles. Networks, usability, and security lead toward support, administration, and security-minded design. The entrepreneurship unit is the version of that work in which you, rather than an employer, choose the user. After HSC, the usual academic doors are bachelor’s programmes in computer science, software engineering, information technology, and related data programmes, where the entrance rules are set by the university, not by this lecture. A college laboratory assistant, a junior developer, a data-entry analyst who can actually query, and a teacher are all coherent next steps. None of them requires you to have founded a company at 17. The MVP is practice at honesty about evidence.

EXAM TIP

A five-mark entrepreneurship question wants the beachhead, the MVP, the riskiest assumption, and the failure rule. A definition of “startup” with no user is the answer that loses the marks. Mention one ethical limit if the offer could easily become copied material or a pile of personal data.

Key terms

Term	Meaning in this lecture
Prototype	A model for learning whether people can use a design.
MVP	The smallest real offer that tests whether the business assumption holds.
Beachhead market	The first narrow group of users you will actually reach.
Riskiest assumption	The belief that would kill the idea if it is false, and that you are least sure of.
Failure rule	The result, chosen in advance, that will make you change or stop.

Check your understanding

Attempt each question before you read the model answer. In the examination, a short question usually earns marks for a definition plus one precise point or a tiny example.

Why is a paper screen not an MVP?

MODEL ANSWER

It can test understanding, which is a prototype question. An MVP must let the user complete the real job well enough that their behaviour tests the business assumption, such as coming back or paying.

Rewrite “all students in Pakistan” as a beachhead.

MODEL ANSWER

One possible beachhead is Class XII CS students in a single college who attend the Tuesday practical. You can reach them this month. You cannot honestly test “all students in Pakistan” with a class project.

Why must the failure rule be written before the test?

MODEL ANSWER

After the test it is easy to move the line until the result looks like success. A rule written first is what makes the test able to fail.

Board-style practice

Multiple choice

Each item is worth one mark. Cross out two options that cannot be right before you choose.

1. The riskiest assumption is the one that

- A. is the easiest to put on a poster
- B. would kill the idea if false, and is still unchecked
- C. the founder likes most
- D. requires the largest logo

2. An MVP differs from a prototype because the MVP

- A. includes every planned feature
- B. is real enough that user behaviour can test the business guess
- C. must be built in Python
- D. cannot be small

3. Which offer is acceptable practical work?

- A. Selling scans of the textbook
- B. An original weekly practice card you wrote and measured
- C. A service that writes other students' practicals
- D. A collected set of identity cards in a public chat

4. A beachhead market should be

- A. as large as possible in the first sentence
- B. narrow enough that you can reach those people and see a failure
- C. secret from the users
- D. defined only by age

Answer key with reasons

Q	Answer	Why
1	B	Risk here means both damage to the idea and lack of evidence. A logo is rarely that assumption.
2	B	The MVP is small on purpose. The difference is that the core job really happens.
3	B	Original work that is tested is the assignment. The other options copy, cheat, or expose people.
4	B	A test needs a group you can actually contact. A continent is not a first test.

Short questions

S1. Distinguish a prototype and an MVP using the practice-card idea.

MODEL ANSWER

A paper sketch of the card, watched while a classmate tries to use it, is a prototype. It asks whether the layout is understandable. Sending a real card two weeks running and counting who completes it is an MVP. It asks whether the behaviour you need actually happens.

S2. Write a failure rule for an SMS revision service aimed at ten classmates.

MODEL ANSWER

Before sending anything, decide that the test fails if fewer than six of the ten answer the mid-week question in both weeks. Replies from the founder do not count. If it fails, do not build an app; change the time of day or stop.

S3. Name one career that uses SQL and one that uses usability tests, as this course described them.

MODEL ANSWER

SQL is daily work in data and information-system roles that answer questions from tables. Usability tests are daily work in design and in software teams that watch a user attempt a task before they call a screen finished. Neither job title is a promise of employment; both are coherent uses of the skills.

Long questions

L1. (5 marks) Choose a problem in your college. Define the beachhead, the MVP, the riskiest assumption, and the failure rule. State one ethical limit.

MODEL ANSWER

Problem: students arrive at the lab without the practical sheet.

Beachhead: students of your own CS section who use the class chat.

MVP: for two weeks, the day before each practical, you post the teacher's own instruction in three lines plus a checklist you wrote. You do not scan a book.

Riskiest assumption: at least half the section will open the message before the practical, which you will see from replies or from a one-tap poll, not from your impression.

Failure rule: if fewer than half respond in both weeks, you stop the posts or you change the time. You do not add features.

Ethical limit: you do not collect phone numbers beyond the chat that already exists, and you do not post anyone's marks or identity card to "personalise" the reminder.

L2. (5 marks) A classmate says their MVP is "an AI app for all Pakistani farmers", with no users yet. Rewrite the plan so it could be tested this month, and explain why the original sentence is not an MVP.

MODEL ANSWER

The original sentence names a technology and a continent-sized market, and it offers nothing a farmer can complete this month. That is a slogan. An MVP has to be usable for the core job.

Rewrite: beachhead is ten vegetable growers the student's family already knows on one road. The riskiest assumption is that they will answer a weekly message about tomorrow's wholesale price at a named market. The MVP is the founder typing those messages from prices they themselves checked, for three weeks. Failure is fewer than six growers replying in week two or week three. There is no model, no app, and no claim of artificial intelligence until that behaviour exists. The ethical limit is that the prices must be ones the founder actually saw, not invented numbers, and the growers can tell them to stop.

Practical work

The practical paper is 25 marks. Keep a journal: aim, steps, the program or diagram, the output, and one sentence on what went wrong the first time.

Lab 6 — An MVP test you can finish

- 1 Write the six lines: beachhead, job, riskiest assumption, MVP, failure rule, ethical limit.
- 2 Build only what that MVP needs. Paper plus a real message counts if the job is a message.
- 3 Run the test on the beachhead, not only on your project partner.
- 4 Record the number you promised to watch, and the decision the rule requires.

WHAT SUCCESS LOOKS LIKE

The journal shows a number chosen in advance and a decision that follows it, including a decision to stop or change.

Copy this summary

- A prototype tests use. An MVP tests the business assumption with a real, minimal offer.
 - A beachhead is a small group you can actually reach.
 - The riskiest assumption kills the idea if it is false. Test that, not the logo.
 - Write the failure rule before you start, and accept the result.
 - Do not sell copied books, ghost-written practicals, or other people's identity documents.
 - The technical units point at development, data, support, security-minded design, and teaching.
- The MVP is practice at evidence.

Revision and the practical journal

Results worth being able to reproduce

Topic	A check you can do in one minute
BST inorder	The sequence is sorted. If it is not, the tree or the walk is wrong.
Preorder / postorder	Node first, or node last, with left before right unless the question says otherwise.
Precision	TP / (TP + FP). In the lecture matrix, 90/120 = 75 percent.
P-value	Surprise if H0 is true. Not the probability that H0 is true.
3NF	No non-key column depends on another non-key column.
File mode w	Replaces the file. Use it when you mean to.
MVP	A real minimal offer plus a failure rule written beforehand.

The lecture tree, for the journal

Root 50; left 30 with children 20 and 40; right 70 with children 60 and 80. Preorder: 50, 30, 20, 40, 70, 60, 80. Inorder: 20, 30, 40, 50, 60, 70, 80. Postorder: 20, 40, 30, 60, 80, 70, 50. Use the question's tree in the examination. Use this one only as a check that your method works.

Practical journal for the 25 marks

- Lab 1: two usability tests and one security control with a recovery path.
- Lab 2: a BST with three traversals and a missing-value search.
- Lab 3: the Vehicle program, a letter count, a 3NF schema, and a checked SQL average.
- Lab 4: precision, recall, accuracy, and a chart you refuse to over-claim.
- Lab 5: a design that names a technology it did not use, and a folder that is not shared by one password.
- Lab 6: an MVP test with a failure rule and the decision that followed from it.

If the practical slot is short, do the tree, the class, the query, and the metric. Those four cover the calculations markers can mark quickly, and your journal still shows the design writing for the rest.